

EEE4114F Final Exam

June 2026

Duration: 3 hours

Total: 80 marks

Empl ID: _____

Instructions:

- This is a closed-book exam.
- There are two parts to this exam.
- **Part A** has 4 questions totalling 40 marks, including 10 marks for multiple-choice questions. You must answer all of them.
- **Part B** has 4 questions totalling 40 marks, including 10 marks for multiple-choice questions. You must answer all of them.
- Both parts can be answered in the space provided on the exam script, and you can write on the back of the printed pages if required — but please specify where you do so. Answer books will be available should you need additional writing space.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs is provided separately.
- You have 3 hours.

Question	Marks	Grade
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	13	
8	7	
Total:	80	

Part A: Digital Signal Processing

Question 1: Multiple Choice [10 marks]

1. What is the z-transform of the signal defined as $x[n] = u[n] - u[n - N]$? (2)

- ☐ $X(z) = \frac{1+z^N}{1-z^{-1}}$
☐ $X(z) = \frac{1-z^{-N}}{1-z^{-1}}$
☐ $X(z) = \frac{1-z^N}{1-z^{-1}}$
☐ $X(z) = \frac{1+z^{-N}}{1-z^{-1}}$

2. What is the 4-point circular convolution of the sequences below? (2)

$$x_1[n] = [2 \ 1 \ 2 \ 1] \quad \text{and} \quad x_2[n] = [1 \ 2 \ 3 \ 4]?$$

- ☐ $[14 \ 14 \ 16 \ 16]$
☐ $[16 \ 16 \ 14 \ 14]$
☐ $[14 \ 16 \ 14 \ 16]$
☐ $[16 \ 14 \ 16 \ 14]$

3. If all the poles of $H(z)$ lie inside the unit circle, then the system must be (2)

- ☐ Causal
☐ Stable and causal
☐ Stable
☐ None of the above

4. What is the output signal when $x(t) = \cos(2\pi 40t)$ is sampled with a sampling frequency of 20 Hz? (2)

- ☐ $\cos(4\pi n)$
☐ $\cos(8\pi n)$
☐ $\cos(2\pi n)$
☐ $\cos(\pi n)$

5. An LTI system is characterised by the system function (2)

$$H(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{2}{1 - 3z^{-1}}.$$

If the system is stable, what is its impulse response?

- ☐ $h[n] = (0.5)^n u[n] + 2(3)^n u[-n - 1]$
☐ $h[n] = -(0.5)^n u[-n - 1] - 2(3)^n u[-n - 1]$
☐ $h[n] = (0.5)^n u[n] - 2(3)^n u[-n - 1]$
☐ $h[n] = (0.5)^n u[n] + 2(3)^n u[n]$

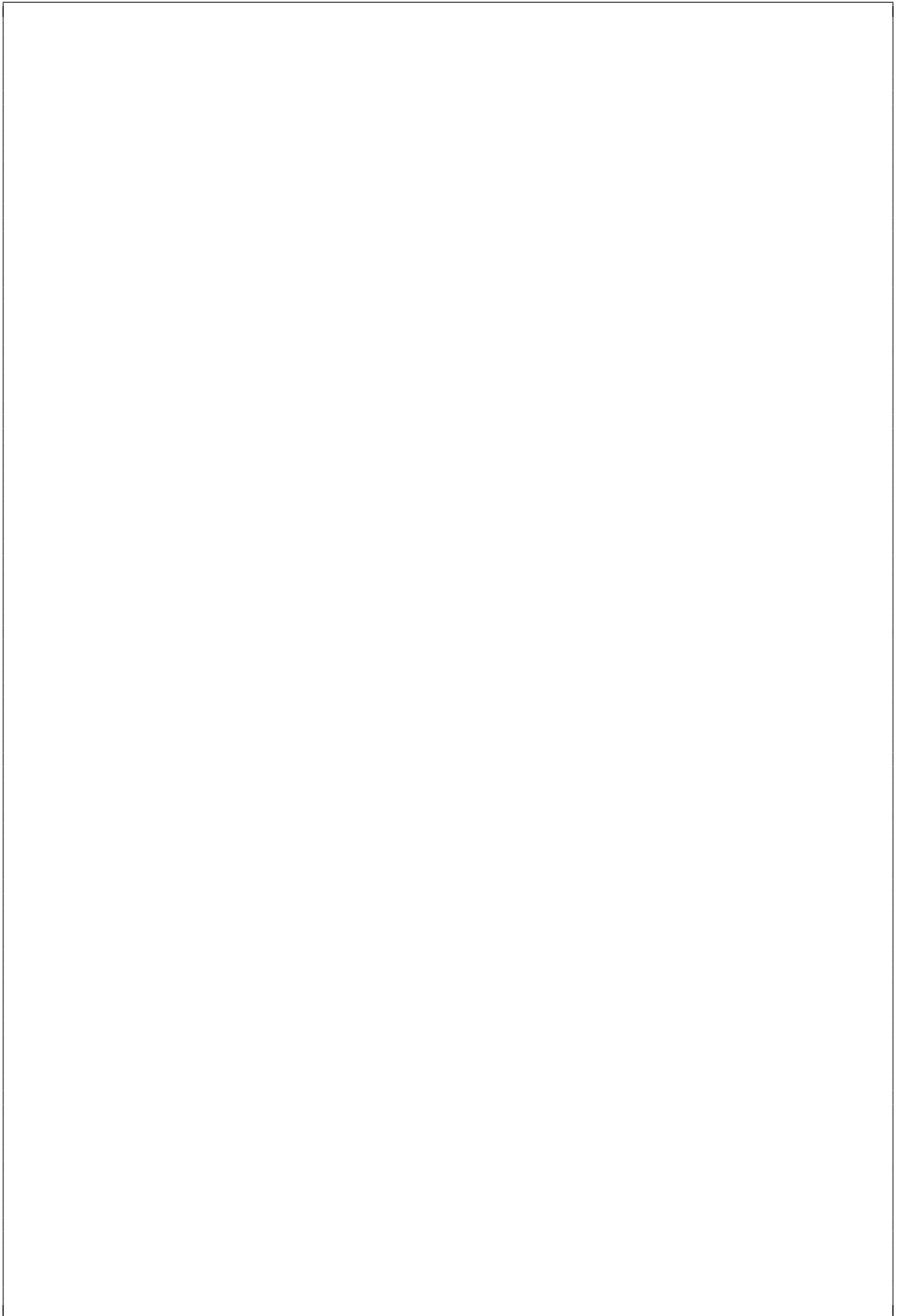
Question 2: Circularity and DFTs [10 marks]

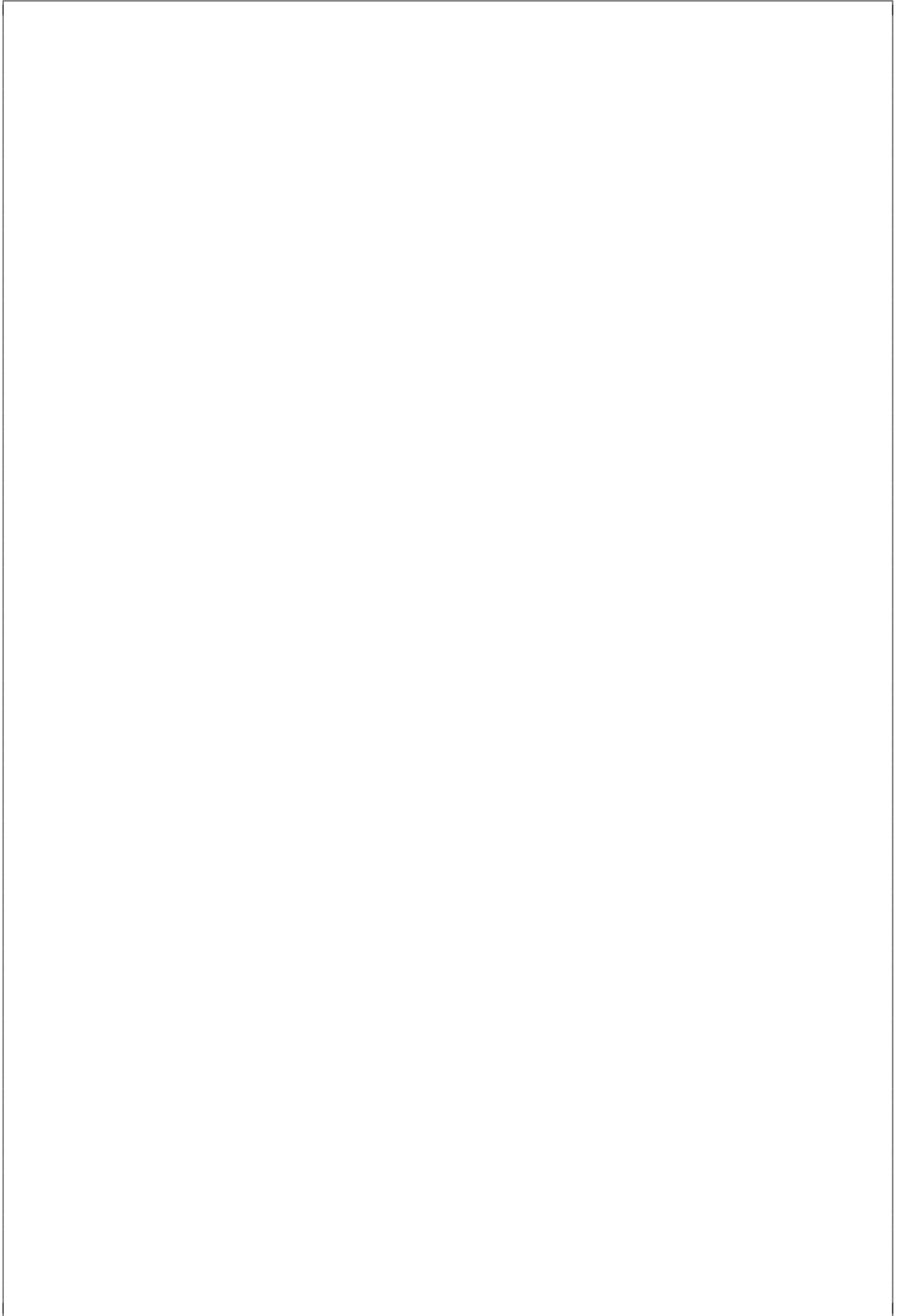
Consider the sequence

$$x[n] = (0.5)^n u[n].$$

Evaluate and sketch each of the following signals over a sensible range:

1. $y[n] = x[(n+2)_3]$ (2)
2. $|X[k]|$, where $X[k]$ is the 3-point DFT of $x[n]$ (4)
3. $x[n] \otimes x[n]$, where $N = 3$ (4)



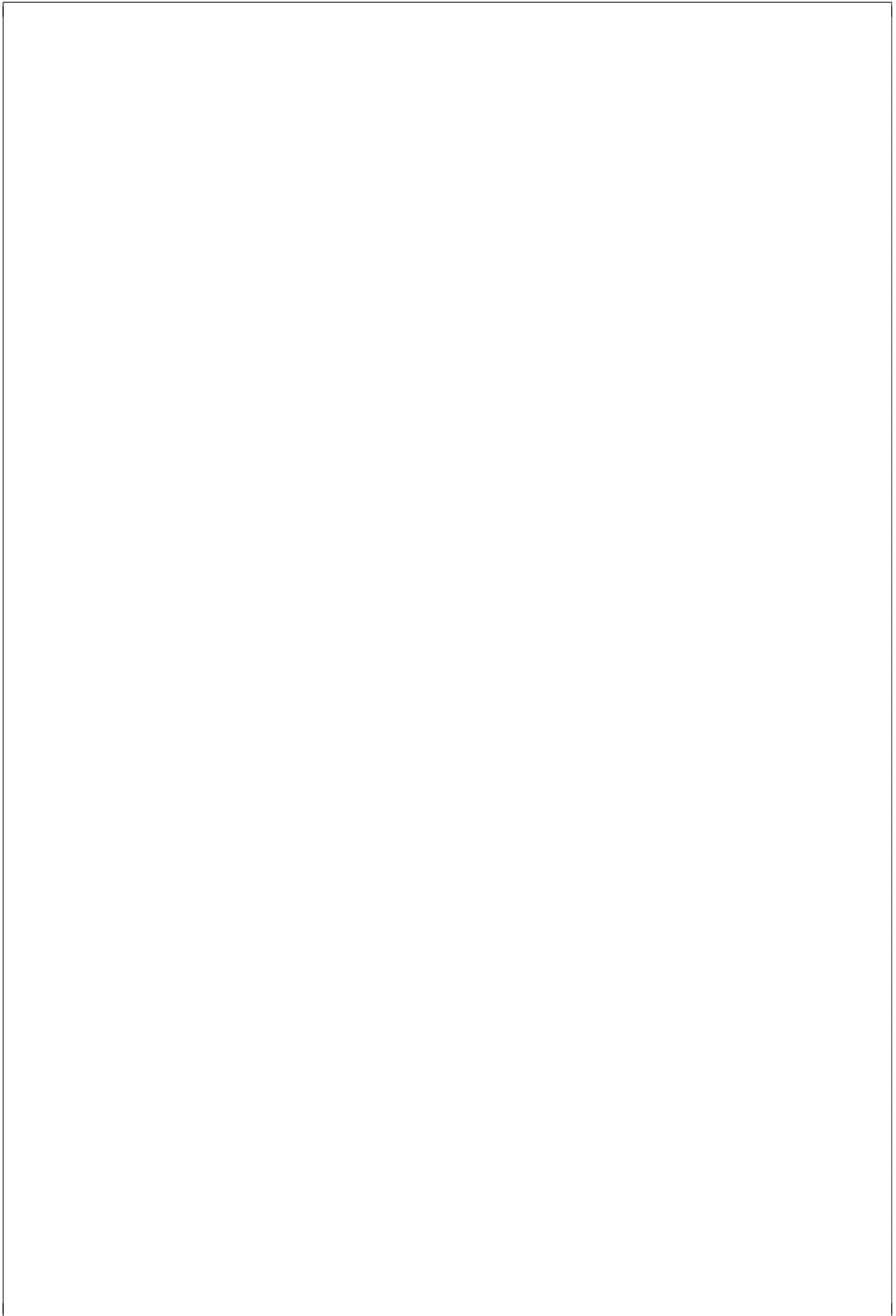


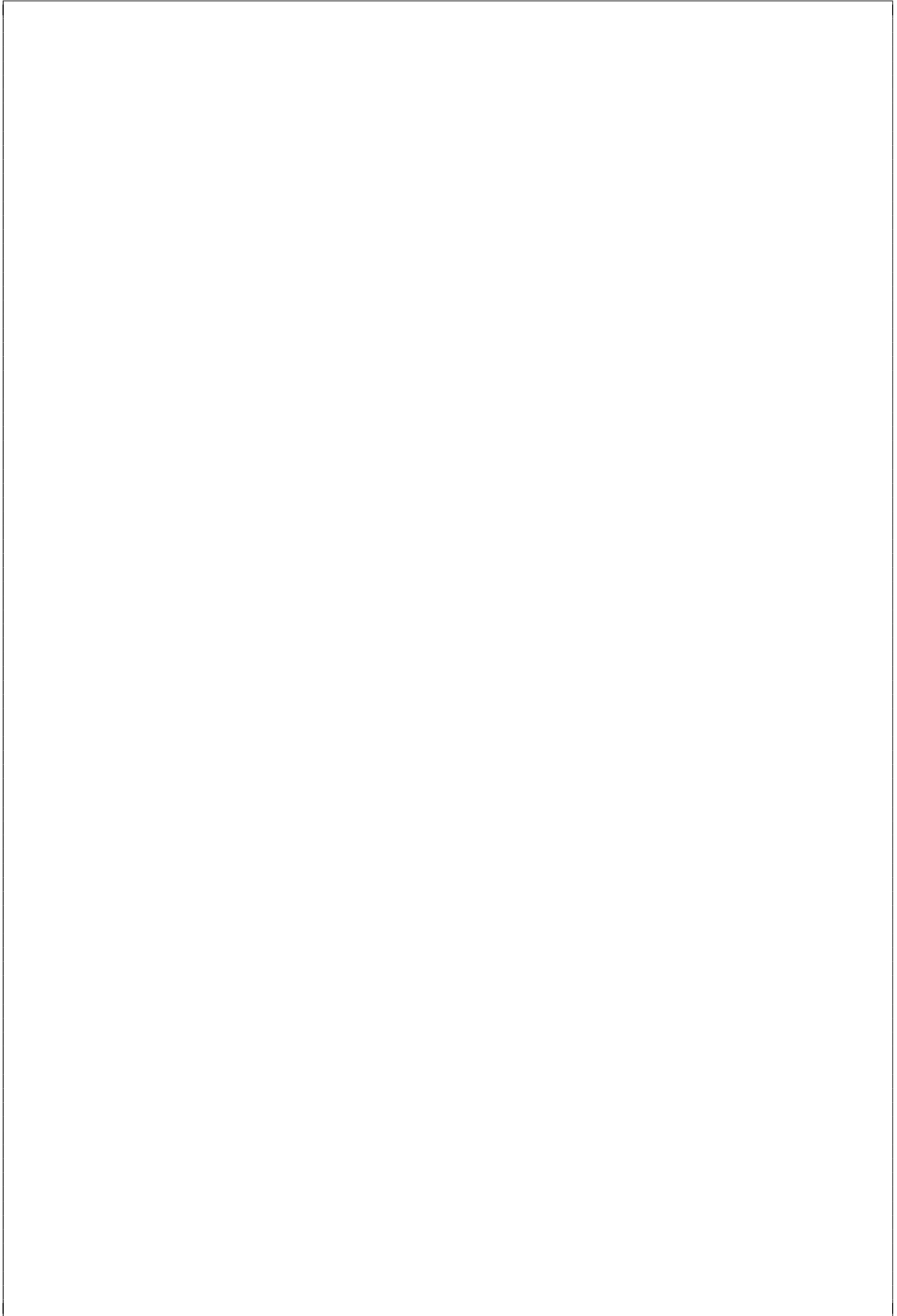
Question 3: Systems and z-transforms [10 marks]

An LTI system has a step response given by

$$y[n] = (2/5)(1/2)^n u[n] + (3/5)(-3/4)^n u[n]$$

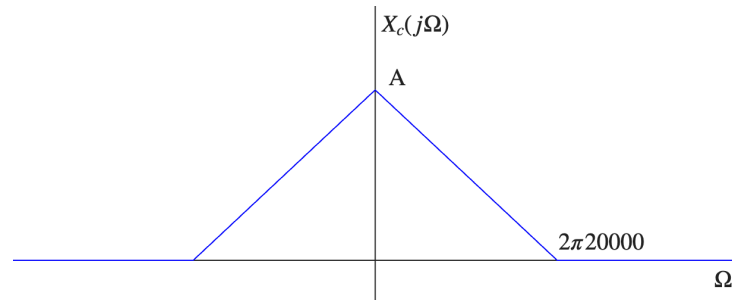
1. Find an expression for the system function, $H(z)$, and determine its ROC. (4)
2. Determine the system's impulse response, $h[n]$. (3)
3. Does the DTFT of $h[n]$ exist? Why? (1)
4. Provide a difference equation that characterises the system. (2)



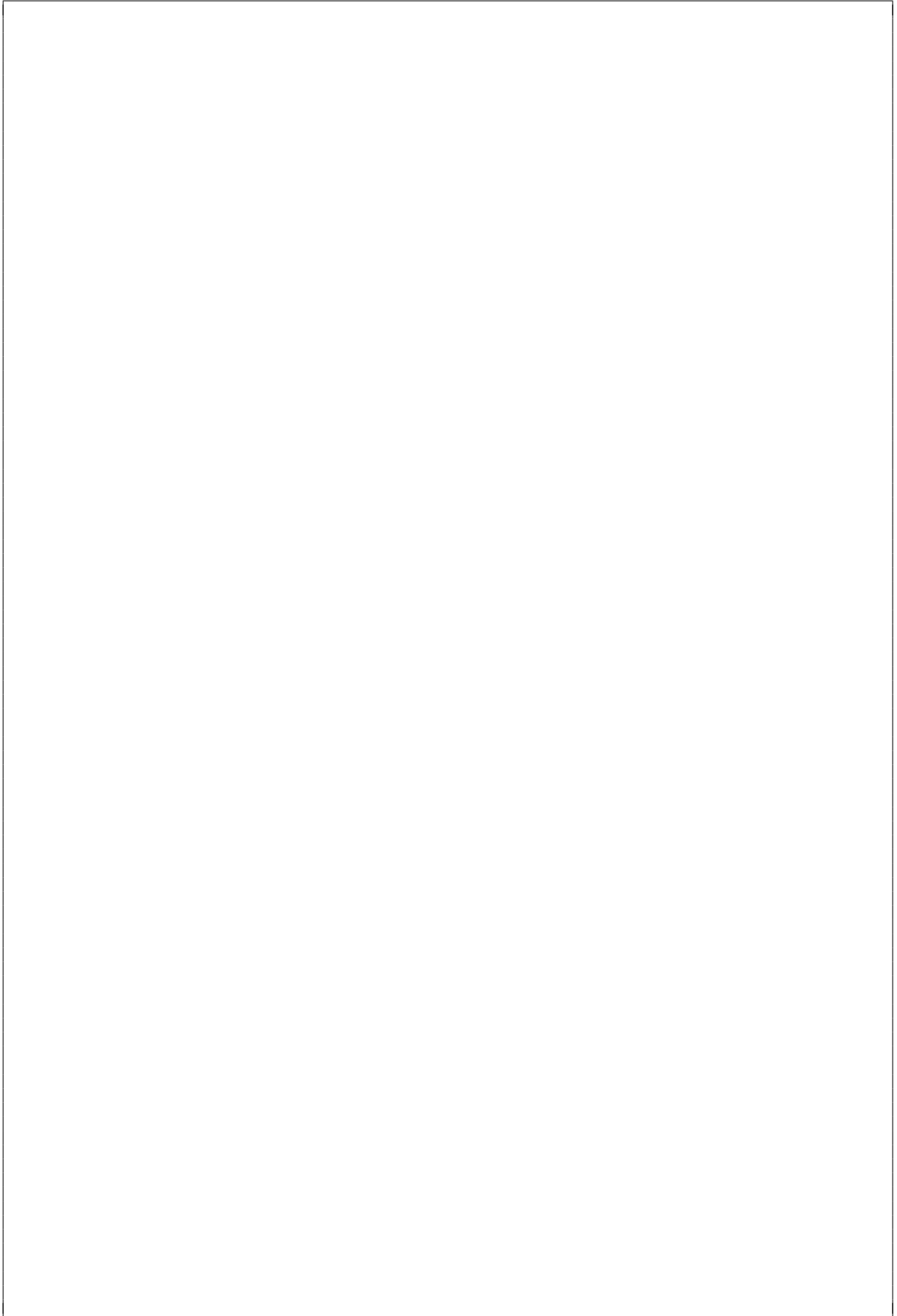


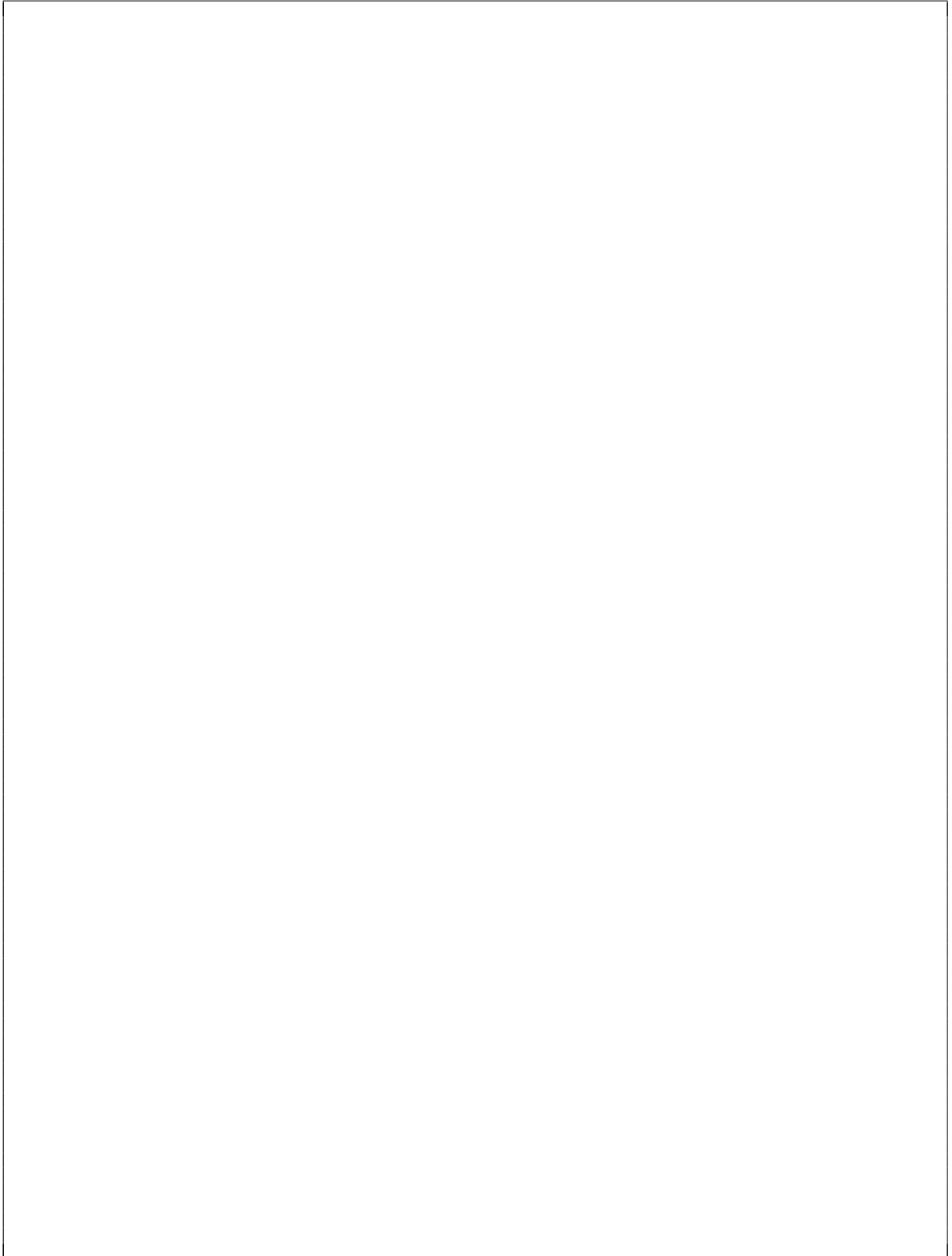
Question 4: Sampling and Frequency Response [10 marks]

We want to use digital filtering methods to process a continuous-time signal $x_c(t)$, which has the following spectrum:



1. Find and sketch the frequency response of an appropriate digital filter, $H(e^{j\omega})$, to remove all frequency content in the input signal over the band 5 kHz to 10 kHz. Assume that a sampling frequency of 60 kHz is used. (5)
2. Suppose we instead decided to implement a digital filter, $H'(e^{j\omega})$, with a stopband from $\pi/4$ to $\pi/2$ rad/sample. Provide a rough sketch of the pole-zero plot for this system in the z -plane, given that the system is both stable and causal, has a stable and causal inverse $G(z)$, and has 4 poles and 4 zeros. You may ignore any constant gain factors, and you do not need to include any values for your poles and zeros. (3)
3. What type of filter characteristic would you expect from the inverse system $G(z)$? Briefly explain. (2)





END OF PART A

Part B: Machine Learning

Question 5: Multiple Choice [10 marks]

1. For a linear regression model, $h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$, the value θ_0 is an example of a (2)
 - ☐ hyperparameter
 - ☐ input feature
 - ☐ model parameter
 - ☐ output feature
2. For a k-Nearest Neighbour algorithm, k is an example of a (2)
 - ☐ hyperparameter
 - ☐ input feature
 - ☐ model parameter
 - ☐ output feature
3. A fully-connect neural network, with one hidden layer has 3 input features, 2 neurons in the hidden layer, and 2 output features. How many model parameters are there (including bias parameters)? (2)
 - ☐ 12
 - ☐ 10
 - ☐ 14
 - ☐ 8
4. If a model exhibits high variance after cross-validation, which of these would be feasible methods for improving performance? You may select multiple answers. (2)
 - ☐ Add regularization
 - ☐ Increase the number of model parameters
 - ☐ Decrease the number of model parameters
 - ☐ Use a repeated three-way holdout method for resampling instead of cross-validation
5. Suppose you have a 2-dimensional input with dimensions of 64×64 . If you had a 2D convolutional neural network that has a convolutional layer with six 4×4 kernels, how many model parameters are there for that layer (assuming that bias terms are included)? (2)
 - ☐ 102
 - ☐ 96
 - ☐ 4198
 - ☐ 4192

Question 6: Fundamental Concepts [10 marks]

Suppose you are given the plot of training data, test data, and a linear model output in Figure 1. The data are split using a single holdout method.

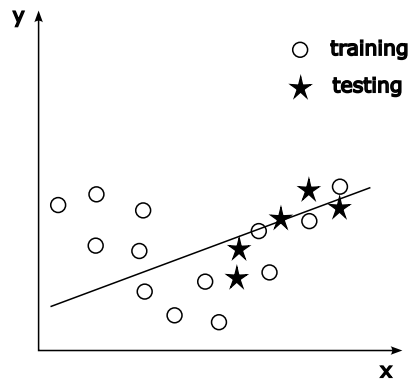


Figure 1: A simple linear regression model. Training data are shown as circles, test data are shown as stars.

1. Would this model appear to generalize well based only on the test data provided? Explain. (2)

2. Would this model be likely to have high or low bias? Why? (2)

3. Would this model be likely to have high or low variance? Why?

(2)

4. Is regularization likely to improve the generalization performance of this model?

(2)

5. Would the use of cross-validation improve the model performance? Explain your answer.

(2)

Question 7: Neural Networks [13 marks]

Suppose you decide to train an artificial neural network to solve a machine learning problem. The hidden layer uses relu activation functions and the output uses a linear activation.

The initial neural network parameters are:

$$\mathbf{W}^1 = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 1 & 5 \end{bmatrix}, \mathbf{W}^2 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \mathbf{b}^1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, b^2 = 0$$

A single input, $\mathbf{x}$, is passed through the network and the corresponding ground truth, y , is provided.

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, y = [21]$$

The following values are determined after the first forward pass:

$$\mathbf{z}^1 = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}, \mathbf{z}^2 = [19]$$

The cost function is the mean square error $J = \frac{1}{2}(\hat{y} - y)^2$ and the model parameters are updated after this single forward pass.

Answer the following questions:

1. Draw a graphical representation of the neural network and label all input features, model parameters, net inputs, and activations. You do not need to include numerical values, just the symbolic representation e.g., x_1 , w_{11}^1 , z_1^1 , a_1^1 , etc. (2)

2. Write out an expression for $\frac{\partial J}{\partial w_{12}^1}$ using chain rule. (2)

3. Calculate the updated value for w_{12}^1 if $x_1 = 1$ and $y = 21$ for the first forward pass. The parameters are updated using a basic gradient descent learning rule with a learning rate of $\eta = 0.1$. (7)

4. If a CNN had no pooling stages would it still be equivariant to translations of the input data? Explain. (2)

Question 8: Reinforcement Learning [7 marks]

Suppose an agent is placed in an environment as shown in Figure 2a, with Q-values in Figure 2b, that has been developed through training an agent in a virtual environment. The agent can take actions from the following action-space $\mathcal{A} = \{\text{North, East, South, West}\}$ which causes it to move in the corresponding direction. These can be abbreviated N,E,S,W, and the position of the Q value within a cell corresponds to the direction of the action. N is $\uparrow$, E is $\rightarrow$, S is $\downarrow$, and W is $\leftarrow$.

s_0, s_2, s_6 , and s_8 are the terminal states. The agent always starts a new trial in s_4 .

S0	S1 -1	S2 +10
S3 +1	S4	S5 -1
S6 -10	S7 +1	S8

(a) The agent's environment. Rewards are shown if they are non-zero.

	0.5	-0.5
-5		

(b) All non-zero Q-values are shown.

Figure 2: The environment and Q-values for a virtual agent

The Q-values are updated according to the following learning rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R(s') + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

where $\alpha = 0.5$ and $\gamma = 0.8$. Answer the following questions.

1. Suppose a new trial is started. If the agent moves East, would it be using a greedy or ϵ -greedy policy? (1)

2. Suppose the agent moves North after moving East from s_4 . (2)

Fill in the updated Q-values onto Figure 2b. Show your working in the box provided as well.

3. Considering these updated values, in which direction would an ϵ -greedy policy prefer to move if the agent starts in s_4 again? (1)

4. Suppose the agent again chooses to move East and then North, and updates its Q-values. Which direction would it move if a greedy policy were used? Show all working. (3)

END OF PART B